Problem Statement

In VLSI and semiconductor engineering, we operate at scales where even a single particle of dust can render an entire chip defective. What if we applied the same mindset beyond silicon wafers to our own surroundings?

Cleanliness in developed nations such as Japan, is not enforced — it is embedded in their mindset. Being India's premier technological institute, we are uniquely positioned to lead not just in innovation, but in responsibility. Let us start at the most visible and manageable level: from keeping aside classroom dusters to now looking at dustbins!

You are appointed as the Technical Head of the **Dustbin Management Committee (DMC)**. Your mission is to bring automation, intelligence, and reliability to waste management of dustbins across campus.

The campus currently uses three types of authorized dustbins, each deployed in different locations. However, a recurring issue persists: Garbage overflow and spill-offs outside the bins, leading to unhygienic conditions.

To tackle this, you propose an automated monitoring system - a simple mobile app where a user captures an image of a dustbin, and an intelligent algorithm determines:

Does DMC need to take action or not? In other words, shall DMC be informed or not?

Below are the three authorized dustbin types used across campus:

<table>
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<tr>
<td>Blue/Green/Cyan and Standing on ground. Top lead may ON or OFF</td>
<td>Metal Cage fully packed dustbin</td>
<td>Metal stand elevated Blue/Green/Cyan dustbins</td>
</tr>
</table>

Objective :

Design an **image-based algorithm** that predicts:

P(intimate DMC) = the probability or confidence with which app should intimate DMC

The model should output: DMC Intimation Decision ∈ {0, 1}

Where:

- 1 → Intimate DMC
- 0 → Do not intimate DMC

We want to intimate DMC when a dustbin is in a spill or overflow condition. Or garbage is lying just near the dustbin but not inside it.

Garbage is defined as non-useful and thrown manmade material objects. Useful manmade objects carefully placed near dustbins or natural flowers, dry fallen tree leaves won't be called as a garbage.

Your system is **dustbin-centric**, not garbage-centric.

- If garbage is present but no dustbin is detected, then DMC does not consider it as their concern.
- Therefore P(intimate DMC) ≈ 0.0

Your model must understand **context**, not just detect garbage or dustbins.
Your algorithm must output a **final binary decision (0 or 1)**.

Dataset Details:

Training Data:

- You may collect data while roaming the campus and checkout various public locations where any of the above 3 types of dustbins are placed
- You may use combination of pretrained models without re-training on campus data (but your test accuracy may suffer)

- Participants are strongly encouraged to collect as much diverse and high-quality dataset as possible from different campus locations. A richer dataset will not only improve model performance but will also be considered for the **Best Dataset Award**.

Test Data (Provided):

- Resolution: **512 × 512**
- Includes challenging edge cases:
  - Garbage near but no dustbin
  - Occluded / partially visible bins
  - Multiple bins in one frame

To ensure fair evaluation and prevent overfitting, the test data is divided as follows:

- Public Set (10 images)
  - Participants will receive feedback during the competition
- Hidden Test Set (200 images)
  - Used for the final evaluation
  - Final rankings will be based on performance on this set

Starter Code (Optional):

We are providing a basic YOLO-based pipeline to help you get started with dustbin detection.

File: `starter_yolo_detection.py`

The sample images used in this file are NOT provided.

Participants are expected to:

- Collect their own dataset (recommended: campus/public areas)
- Or use publicly available datasets

Expected output:

Participants are required to submit their solution in the following format:

- The submission must be a **single ZIP file** named exactly as your **team name**.
- Inside the ZIP file, there must be **exactly three files**:
  - **model.pkl** (Pickle file containing the trained model)
  - **predict.py** (script used to run inference using your model)
  - **`requirements.txt`** (File listing all necessary dependencies required to run your model)

The **predict.py** file defines how your model will be used during evaluation. It must include the following two functions:

- `load_model()` → Loads and returns your trained model
- `predict(model, image_path)` → Takes an image path as input and returns a **single prediction (0 or 1)**

A **sample `predict.py` has also been provided for reference**, and participants are encouraged to follow a similar format.

This setup allows flexibility to:

- Use any framework (PyTorch, TensorFlow, etc.)
- Apply custom preprocessing (resizing, normalization, etc.)
- Design your own inference pipeline

The submitted **model.pkl** should be a trained model capable of performing inference on unseen images.

Your algorithm must output a **final binary decision (0 or 1)** instead of a probability score

Additionally, please ensure that your dataset strictly follows the required format:

- All input images must be of size **512 × 512 pixels**.

Any deviation from the above format may lead to disqualification or rejection of the submission

Please **do not use GPU while loading the model**, as it may create compatibility issues during evaluation. Avoid saving/loading models in a way that depends on CUDA.

Reason: While you are free to train your models using GPUs, the evaluation environment will run entirely on **CPU**. Heavy GPU dependencies are not required and may lead to execution failures.

Important Guidelines

- Ensure your code runs without errors in a **clean environment**
- Do not use **hardcoded system-specific paths**
- Your submission must be **self-contained within the ZIP folder**
- The evaluator will strictly call only:
  - `load_model()`
  - `predict(model, image_path)`

Disqualification Criteria

Submissions will be rejected if:

- Folder/ZIP naming is incorrect
- `predict.py` is missing or incorrectly implemented
- Required functions are not defined properly
- The code crashes during execution
- Output format is not strictly **0 or 1**

Submission & Evaluation Links:

- **Submission Form:** https://forms.gle/YJb6qnAigpwkP3T49
- **Live Leaderboard:** https://akshat221005.github.io/dmc-leaderboard

Evaluation Criteria

The evaluation of submissions will be primarily based on the performance of your model on the test dataset using the following scoring function:

$$\text{Score} = \frac{1}{2}\sum_{i=1}^{N}(1 - |\hat{y}_i - y_i|)$$

where:

- $\hat{y}_i$ ∈ **{0, 1}** is your model's predicted label
- y_i ∈ **{0, 1}** is the ground truth label
- **N** is the number of samples (e.g. 200 for hidden test set)

Since predictions are strictly binary, this metric effectively rewards correct classifications. A correct prediction contributes **1** to the score, while an incorrect prediction contributes **0**. Therefore, **higher scores correspond to better performance and higher ranking on the leaderboard**.

The overall evaluation is divided as follows:

1. Primary Evaluation (80%)

Based on **model performance**

- Accuracy (probability quality)
- Handling of edge cases

2. Short PPT submission (20%)

- Clarity of approach
- Innovation and intuition
- Explanation of edge case handling
- Model design + improvements

This is to be submitted only by the **top 20 teams on the leaderboard**. The shortlisted teams will be informed after the **10th April**.

3. Bonus (Extra 20%)

Build a **working application interface** where:

- User can:
    - Capture an image **OR**
    - Upload an image
- Your deployed model:
    - Processes the image
    - Outputs spill probability or DMC informing probability
    - Displays:
        - “No Action Needed”
        - “DMC Action Required”

Some examples and output with remarks.

Remake is given just for clarification. Your algorithm should output only 0 to 1 score as probability of informing DMC.

Test Case	Expected Output	Remark
	1.0	100% spillover dustbin, need to auto-inform dustbin cleaning authority (not general garbage collectors)
	**0.0**	No IITK dustbin is installed in the picture. There is no point of informing DMC as they will not take any action. They will take action only when there is dustbin which they have installed (one of 3 types as given in PS)

	**0.0**	No garbage overspill or random garbage nearby those dustbins, so no need to inform DMC.
	**0.0**	It is postbox and not a dustbin. Even if there would have been garbage near this postbox, DMC will not take any action. So all these cases output expected as 0.
	**0.0**	It is just a poster and not real dustbin. DMC will not take any action, P(intimate DMC) = 0
	**1.0**	Institute dustbins with garbage nearby and not inside dustbin, so need to inform DMC

	**1.0**	Garbage all outside, spill over, so call DMC
	**0.0**	This is not one of the authorized dustbins, so there is no need to call DMC.
	**0.0**	Although the dustbin is full, there is no spillover. Therefore, no action from DMC is required.

	**0.0**	Dustbin detected with no garbage spillover. Surrounding leaves are natural litter and not considered waste - DMC action not required.
	**0.0**	Although the lid is on the ground, it is not considered a spill condition. The dustbin shows no overflow or surrounding garbage. Therefore, no DMC action is required.